Application Note

Document No.: AN1193

Quick Start Guide for APM32E10x_SDK

Version: V1.0

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1. Introduction

APM32E10x SDK (Software Development Kit) is a comprehensive set of development drivers and documents designed specifically for the APM32E10x series MCUs. It includes a low-level peripheral driver library, a collection of examples for different development boards, and a rich set of middleware. The SDK aims to significantly improve developer productivity by reducing development workload, time, and cost.

The materials referenced in this application note are:

AN1080

AN1093

Note: These materials can be obtained from the official website www.geehy.com.

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2. Content about SDK

The complete "SDK" directory is as follows:

Figure 1 "SDK" Directory



3. Content about Boards

The "Boards" folder contains the BSP (Board Support Package) for the APM32E103 development board. It can help to quickly drive the peripheral circuits or components on the board. The BSP can be found in the ~/Boards directory.

The provided BSP is built for the APM32E10x mainboard. For other user-developed boards, some minor modifications may be required.

The directory structure of the " APM32E10x SDK Boards" folder is as follows:

- "Board_ APM32E10x_EVAL" folder
- "Board_ APM32E10x_MINI" folder

4. Content about Documents

The "Documents" folder contains a link file that can redirect to Geehy Semiconductor's Technical Support Center. The library support documentation can be found in the ~/Documents directory. This file can be used to view all supported function explanations, parameter functions, and return values.

The directory structure of the "APM32E10x SDK Documents" folder is as follows:

- DATASHEET.pdf
- APM32E10x_um.chm
- 快速上手指南.pdf
- Quick Start Guide.pdf

5. Content about Examples

Example applications can be found in the ～/Examples directory.

The provided examples are built for the APM32E103 EVAL、APM32E103 MINI development board. For use on other user-developed boards, appropriate adjustments are required.

The directory structure of the " APM32E103 SDK Example" folder is as follows:

- Include
- Project
  - Eclipse
  - IAR
  - MDK
- Source
- Clean Project.bat
- readme.txt

All example applications are tested using APM32E103 SDK V1.0.0, including the following examples:

Table 1 List of Example Programs Supported by the Development Board

IP / Module	Example	APM32E103 EVAL	APM32E103 MINI	NA	NA
ADC	ADC_AnalogWatchdog		√		
	ADC_ContinuousConversion		√		
	ADC_DMA		√		
	ADC_DualRegulSimulMode		√		
	ADC_InjectionRegularSampling		√		
	ADC_MultiChannelScan		√		
	ADC_Potentiometer	√			
BAKPR	BAKPR_Tamper		√		
Basic	SWO_Output	√	√		
	Template	√	√		
CAN	CAN_LoopBack		√		

IP / Module	Example	APM32E103 EVAL	APM32E103 MINI	NA	NA
	CAN_Normal		√		
	CAN_Dual	√			
CRC	CRC_Calculation		√		
DAC	DAC_ADC		√		
	DAC_NoiseWave		√		
DMA	DMA_ADCMultiChannel		√		
	DMA_MemoryToMemory		√		
	DMA_UsartToMemory		√		
EINT	EINT_Config		√		
EMMC	DMC_SDRAM	√	√		
FMC	FMC_Program		√		
	FMC_Protection		√		
FPU	FPU_Math		√		
GPIO	GPIO_Toggle		√		
I2C	I2C_TwoBoards		√		
	I2C_EEPROM	√			
I2S_Interrupt	I2S_Interrupt		√		
IAP	IAP_Application1		√		
	IAP_Application2		√		
	IAP_BootLoader		√		
IWDT	IWDT_Reset		√		
NVIC	NVIC_Priority		√		
	NVIC_WFI		√		
PMU	PMU_Standby		√		
	PMU_Stop		√		
RCM	RCM_ClockConfig		√		

IP / Module	Example	APM32E103 EVAL	APM32E103 MINI	NA	NA
RTC	RTC_Alarm		√		
	RTC_Second		√		
	RTC_Clock	√			
RTOS	CMSIS_RTX		√		
	FreeRTOS		√		
	RT-thread		√		
SCI2C	SCI2C_TwoBoards		√		
SDIO	SDIO_SDCard	√	√		
SPI	SPI_FullDuplex		√		
	SPI_Flash	√			
	SPI_LCD	√			
SysTick	SysTick_TimeBase		√		
TMR	TMR_6Steps		√		
	TMR_32BitCount		√		
	TMR_CascadeSynchro		√		
	TMR_EncoderInterface		√		
	TMR_ExtTriggerSynchro		√		
	TMR_OCActive		√		
	TMR_OCInactive		√		
	TMR_OCToggle		√		
	TMR_PWMInput		√		
	TMR_PWMOutput		√		
	TMR_SinglePulse		√		
	TMR_TimeBase		√		
	TMR_TMR1DMA		√		
	TMR_TMR1DMABurst		√		

IP / Module	Example	APM32E103 EVAL	APM32E103 MINI	NA	NA
USART	USART_Interrupt		√		
	USART_IrDA		√		
	USART_LIN		√		
	USART_Printf	√	√		
	USART_Smartcard		√		
USB	USBD_CDC	√	√		
	USBD_Composite_CDC	√	√		
	USBD_Composite_CDC_HID	√	√		
	USBD_Composite_CDC_MSC	√	√		
	USBD_Composite_CDC_WINUSB	√	√		
	USBD_Composite_HID_MSC	√	√		
	USBD_Composite_HID_WINUSB	√	√		
	USBD_Composite_MSC_WINUSB	√	√		
	USBD_Composite_WINUSB	√	√		
	USBD_CUSTOM_HID	√	√		
	USBD_CUSTOM_HID_Keyboard	√	√		
	USBD_HID	√	√		
	USBD_HID_WakeUp_LowPower	√	√		
	USBD_MSC	√	√		
	USBD_MSC_LowPower	√	√		
	USBD_WINUSB	√	√		
WWDT	WWDT_Reset		√		

6. Content about Libraries

The "Libraries" folder contains a series of library files. It can provide support for APM32E10x MCU, such as device support, startup files, linker files, and Std peripheral support. The library files can be found in the ～/Libraries directory.

The "APM32E10x SDK Libraries" folder includes the following materials:

- APM32E10x_Driver
- CMSIS
- Device

6.1. APM32E10x_StdPeriphDriver

This folder contains all Std peripheral library driver files.

Table 2 List of Std Drivers Supported by the Device

IP / Module	APM32E103	NA	NA	NA
apm32e10x_adc	√			
apm32e10x_bakpr	√			
apm32e10x_can	√			
apm32e10x_crc	√			
apm32e10x_dac	√			
apm32e10x_dbgmcu	√			
apm32e10x_dma	√			
apm32e10x_dmc	√			
apm32e10x_eint	√			
apm32e10x_fmc	√			
apm32e10x_gpio	√			
apm32e10x_i2c	√			
apm32e10x_iwdt	√			
apm32e10x_misc	√			
apm32e10x_pmu	√			
apm32e10x_rcm	√			

IP / Module	APM32E103	NA	NA	NA
apm32e10x_rtc	√			
apm32e10x_sci2c	√			
apm32e10x_sdio	√			
apm32e10x_smc	√			
apm32e10x_spi	√			
apm32e10x_tmr	√			
apm32e10x_usart	√			
apm32e10x_usb	√			
apm32e10x_usb_device	√			
apm32e10x_usb_host	√			
apm32e10x_wwdt	√			

6.1.1. Std peripheral library

The Std peripheral library provides peripheral development interfaces in the form of regular functions.

Regular functions are declared in the header files of the Std peripheral library and implemented in the source files. They are responsible for operations such as peripheral parameter configuration, function control, data transmission and reception, status reading, as well as interrupt and flag management.

The common naming characteristics of regular functions are shown in the following table:

Table 3 Naming Characteristics of Std Inline Functions

Scenario	Function	Naming Characteristic
General configuration	Peripheral/module configuration	PPP_Config
	Subfunction configuration	PPP_ConfigXXX
	Structure initialization	PPP_ConfigStructInit
	Default initialization of a configuration	PPP_ConfigXXXStructInit
Reset	Peripheral/module reset	PPP_Reset
	Subfunction reset/clear	PPP_ResetXXX / PPP_XXXReset
Enable control	Enable a peripheral/module	PPP_Enable
	Disable a peripheral/module	PPP_Disable

	Enable a subfunction	PPP_EnableXXX
	Disable a subfunction	PPP_DisableXXX
Event generation	Generate an event directly	PPP_GenerateXXX
	Enable generation of a condition/signal	PPP_EnableGenerateXXX
	Disable generation of a condition/signal	PPP_DisableGenerateXXX
Flag	Read a status flag	PPP_ReadStatusFlag
	Clear a status flag	PPP_ClearStatusFlag
Interrupt	Enable an interrupt	PPP_EnableInterrupt
	Disable an interrupt	PPP_DisableInterrupt
	Read an interrupt flag	PPP_ReadIntFlag
	Clear an interrupt flag	PPP_ClearIntFlag
DMA	Enable DMA	PPP_EnableDMA
	Disable DMA	PPP_DisableDMA
	Configure DMA	PPP_ConfigDMA
	Enable a DMA subfunction	PPP_EnableDMAXXX
	Disable a DMA subfunction	PPP_DisableDMAXXX

6.2. CMSIS

This folder contains header files related to the Arm® Cortex®-M3 core, which define the registers and functions of the core.

6.3. Device

This folder contains device-specific files related to the APM32E10x. Wherein:

- The "Geehy\APM32E10x\Include" folder contains all APM32E10x device header files, which define the device registers and interrupt vectors.
- The "Geehy\ APM32E10x \Source" folder contains startup files and linker files for the MDK, IAR, and GCC platforms, including the reset vector table and startup code.

7. Content about Middlewares

The "Middlewares" folder contains a series of third-party middleware. Such middleware can be found in the ~/Middlewares directory.

The middleware used in the "APM32E10x SDK Middlewares" folder includes the following materials:

- USB_Library
- RTThread
- lwip-1.4.1
- FreeRTOS
- fat_fs

7.1. USB_Library

The USB_Library folder contains the USB development framework.

7.2. RTThread

The RTThread folder contains the RT-Thread Nano real-time operating system.

7.3. lwip-1.4.1

The lwip-1.4.1 folder contains the lwIP lightweight TCP/IP protocol stack.

7.4. FreeRTOS

The FreeRTOS folder contains the FreeRTOS operating system based on the CMSIS RTOS2 standard.

7.5. fat_fs

The fat_fs folder contains the fat_fs file system.

8. Content about Package

The "Package" folder includes the Geehy APM32E10x DFP Package. This package can be found in the ~/Package directory.

The " APM32E10x DAL SDK Libraries" folder includes the following materials:

- Geehy. APM32E1xx_DFP.1.0.5.pack

8.1. SVD

This folder contains SVD files, which allow users to simulate and view register usage.

8.2. Pack Package

This folder stores the latest version of the Keil pack package, for use in the MDK-ARM environment to support the APM32E10x series chips. The latest version of the Keil pack package can also be obtained from the following link:

https://www.keil.arm.com/packs/

https://www.geehy.com/design/software

9. Content about Utilities

The "Utilities" folder includes some utilities. These utilities can be found in the ~/Utilities directory.

The " APM32E10x SDK Utilities" folder includes the following materials:

- FLM
- IAR_AddOn
- pyOCD
- SVD

9.1. FLM

This folder stores the FLM files.

9.2. IAR_AddOn

This folder stores the IAR AddOn exe files

9.3. pyOCD

This folder stores the python script files for pyOCD.

9.4. SVD

This folder contains SVD files, which allow users to simulate and view register usage.

10. Quick Start

To quickly EVBuate the APM32E10x SDK, you may need to prepare the following environment or content:

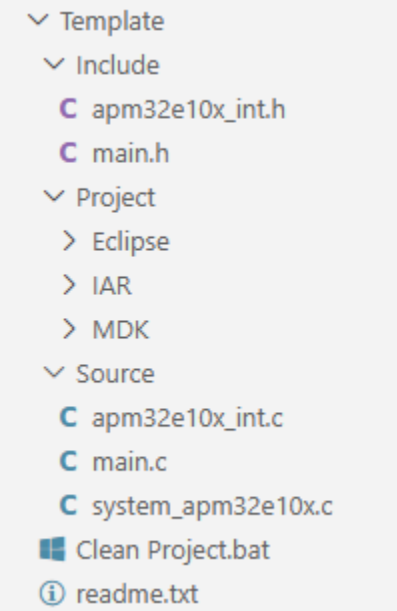
- Window 10/11
- MDK-ARM v5.29 or updated version
- IAR EW for ARM 8.50.5 or updated version
- Eclipse 4.24 or updated version
- arm-gnu-toolchain 10.3.1 or updated version
- Emulate debugger (e.g. Geehy-Link or J-Link)
- Any of the following development boards (depending on the chip used in the project):
  - APM32E103 EVAL Development Board
  - APM32E103 MINI Development Board

Note: The content of this section takes the APM32E103 EVAL Development Board as an example.

10.1. Template Project

The template project is stored in ~/Examples/Board_ APM32E103_MINI/Basic, and its contents are shown in Figure 2. It includes template projects for MDK, IAR, and Eclipse. The "Config" folder contains the initialization code for the devices used, while the "Include" and "Source" folders store the header files and source files used by the application code in the template project, respectively.

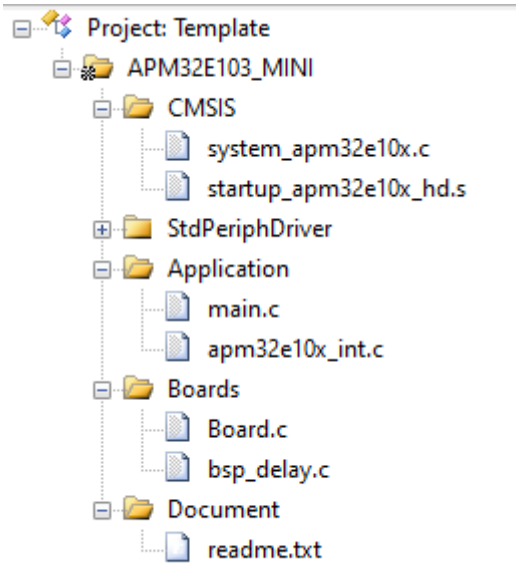
Figure 2 Contents of the Template Folder



Open the folder of the corresponding project and click the project file to open the IDE project. Shown in Figure 3 is the MDK template project. The files in the project are described as follows:

1. system_ apm32e10x.c – System initialization configuration file.
2. startup_ apm32e10x.s – Startup file.
3. apm32e10x_ppp.c in the "StdPeriphDriver" folder: Low-level driver files for various peripherals.
4. Board.c – Onboard support file.
5. apm32e10x_int.c – Interrupt file, containing kernel and peripheral interrupt service functions.
6. main.c – Main source code file of the template project.
7. readme.txt – Project description file.

Figure 3 MDK Template Project



10.2. SDK-related Macro Definitions

When creating a new project, in addition to importing the startup file (startup_ APM32E10x.s) into the project, it is also necessary to enable the relevant macro definitions according to the MCU model. The relationships of the macro definitions used in the SDK are shown in the table below.

Table 4 Macro Definition Relationships

Macro definition	Description
APM32E103_EVAL、APM32E103_MINI	Enable BSP support for the APM32E10x EVAL and MINI development board

10.3. Device Initialization

When initializing peripherals with the standard peripheral library, the peripheral initialization configuration is usually completed directly in `main.c`. In general, the standard peripheral library defines peripheral initialization structure variables in the application. After completing the system clock and basic hardware initialization, each member of the structure is assigned one by one, and then the corresponding initialization configuration interface is called to complete the peripheral initialization.

For example, when initializing USART, a configuration structure variable of type `USART_Config_T` is usually defined first, and then parameters such as baud rate, hardware flow control, operating mode, parity, stop bits, and word length are assigned.

After the parameter configuration is completed, the peripheral initialization interface is called to initialize the USART so that the configuration takes effect.

The USART initialization process should usually include system clock configuration, related

GPIO configuration, USART parameter configuration, and peripheral enabling.

When the transmit function is required, the transmit buffer status flag should also be checked before sending data, and the data transmit interface should be called to complete data writing.

10.4. Compilation and Download

For installation and usage of IDEs and simulation/download toolchains, refer to the following documents:

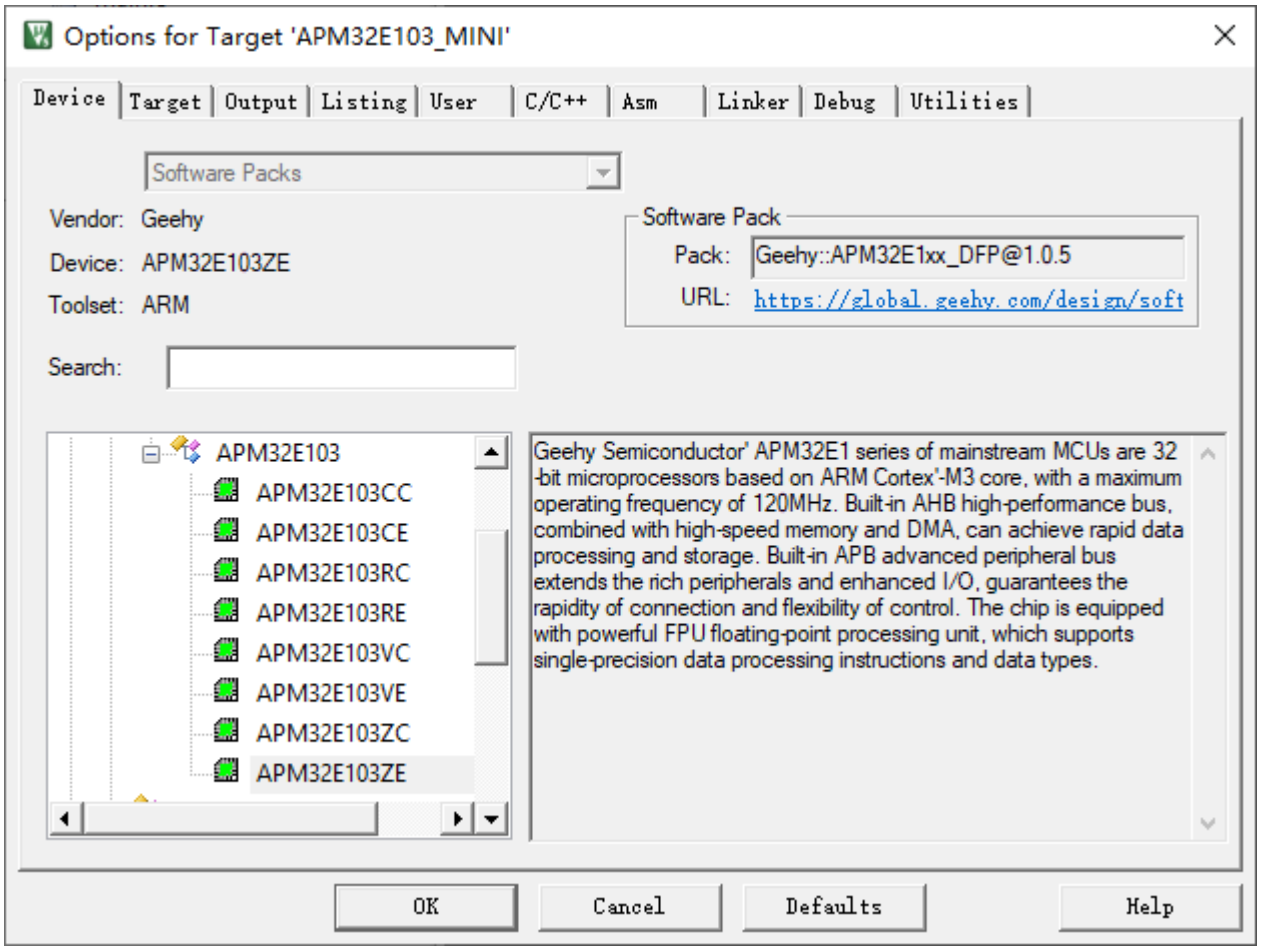
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10.4.1. Configuration Project

After installing the Keil pack toolchain as described in the above documents, open the Template MDK project and click the “Magic Wand” to confirm the chip selection. As shown in Figure 4, select APM32E103ZE for development.

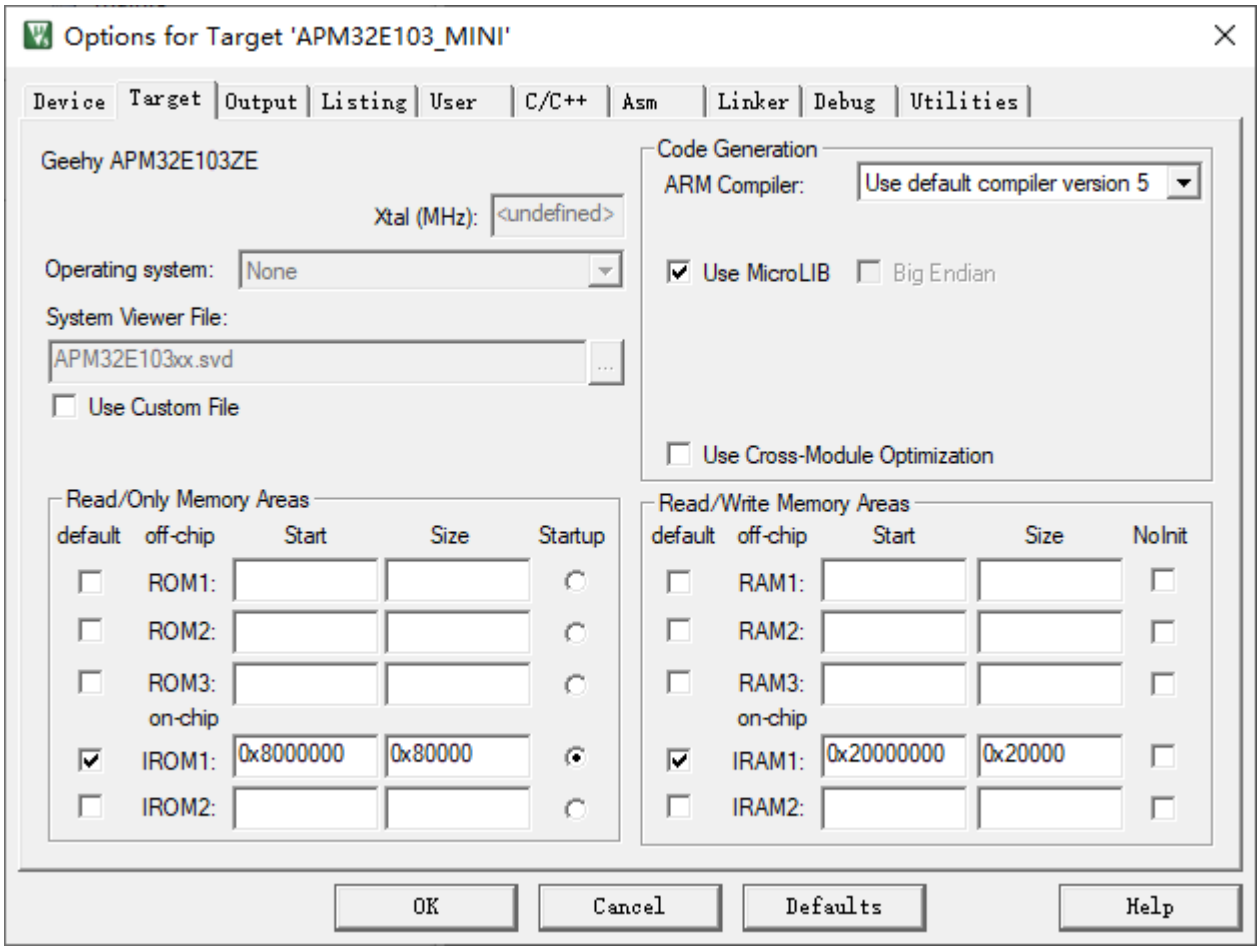
Figure 4 Viewing Chip Selection



By switching to the "Target" tab, you can view the address and size of the ROM and RAM

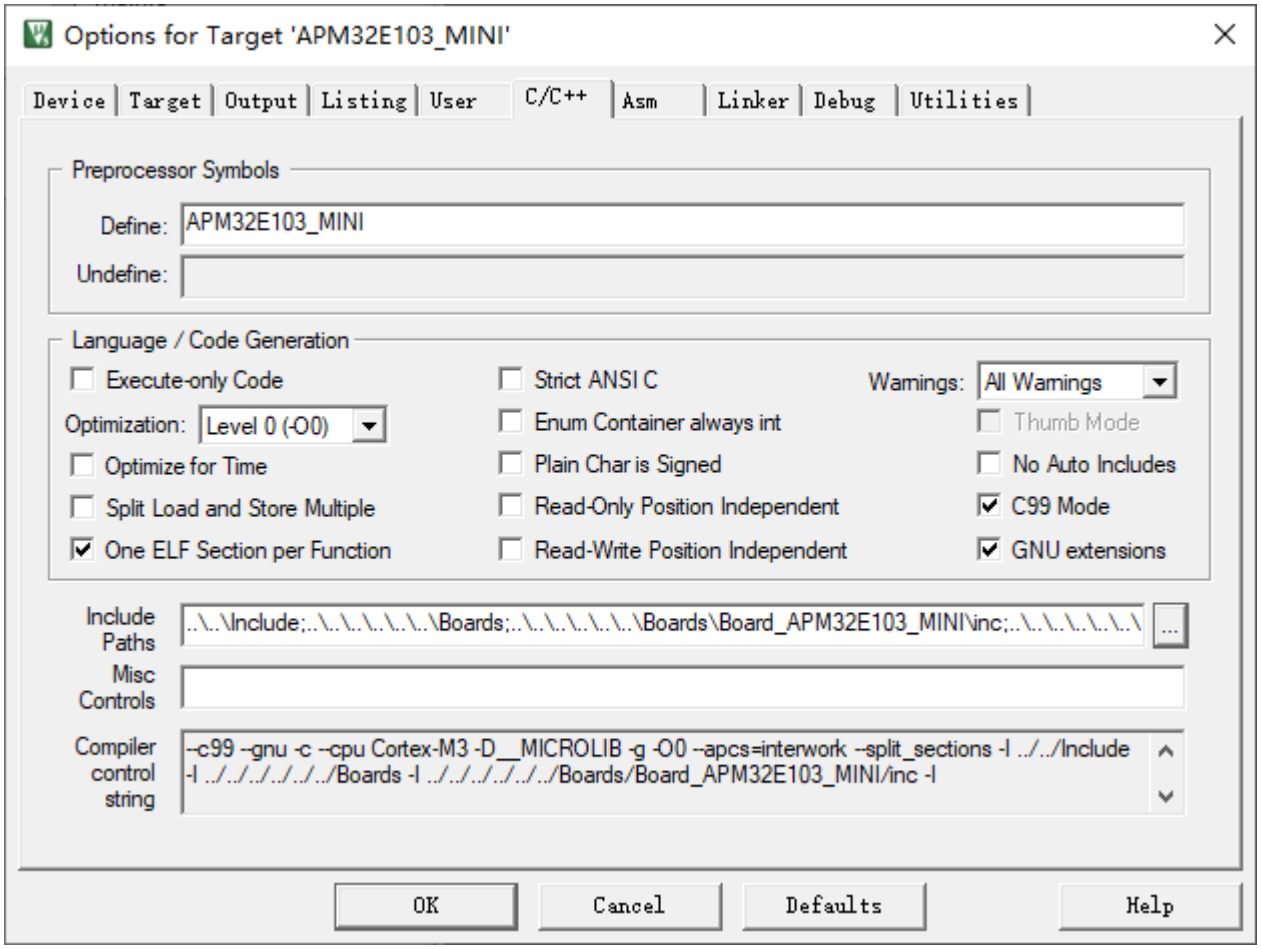
configured for the APM32E103ZE.

Figure 5 ROM and RAM



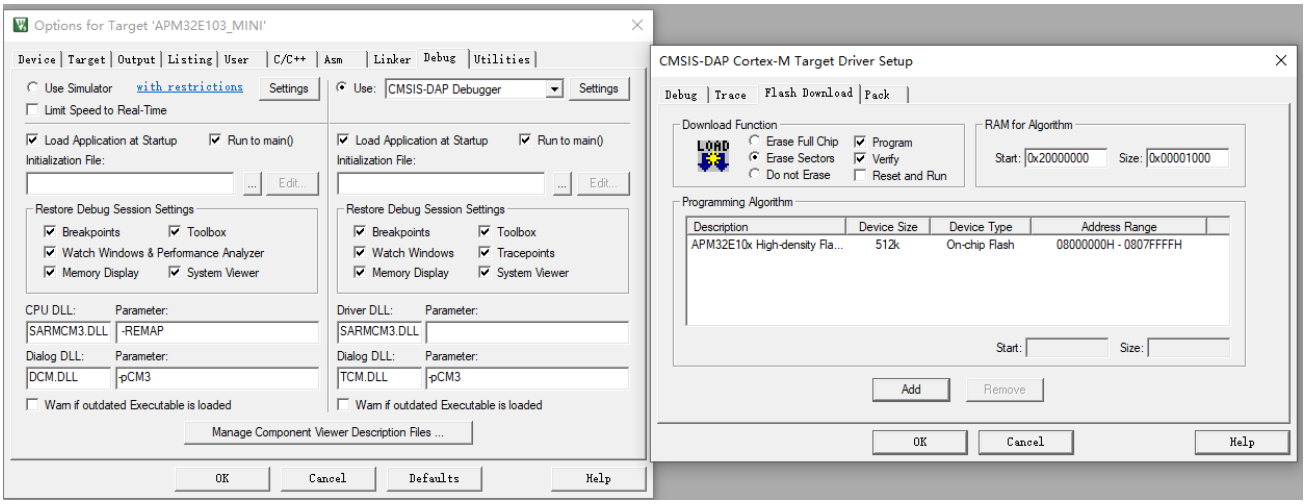
Switch to the C/C++ tab and confirm that the APM32E103 macro definition is enabled to support the development of APM32E103ZE.

Figure 6 C/C++ Configuration



Switch to the "Debug" tab, select the *CMSIS-DAP* Debugger emulator, and click the "Settings" button to enter the "Flash Download" tab. Confirm that the APM32E10x 512KB Flash download algorithm is selected. Note that the algorithm usage range is 0x80000000H ~ 807FFFFF H, as shown in Figure 7.

Figure 7 Debug



10.4.2. Compilation and Download

Click the "Build" button on the MDK interface for compilation. After confirming that the compilation results are correct, click the "Download" button to perform the download. The download result is as shown in Figure 10.

Figure 8 Compilation Project

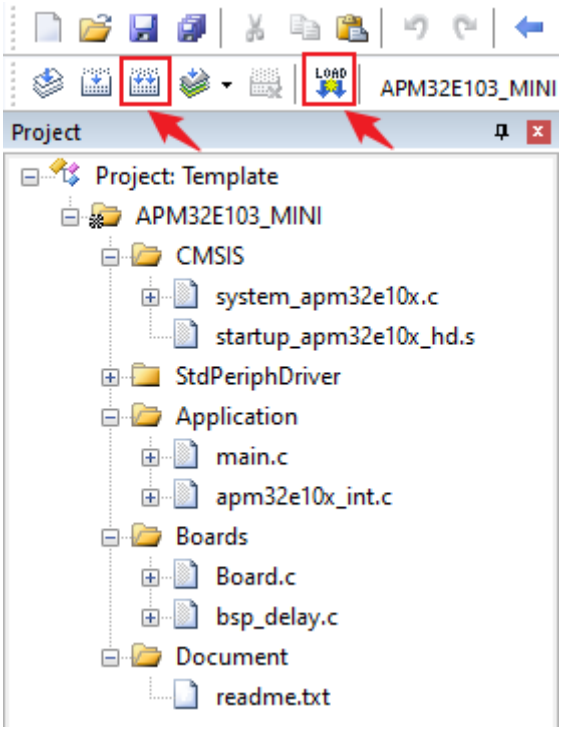


Figure 9 Compilation Results

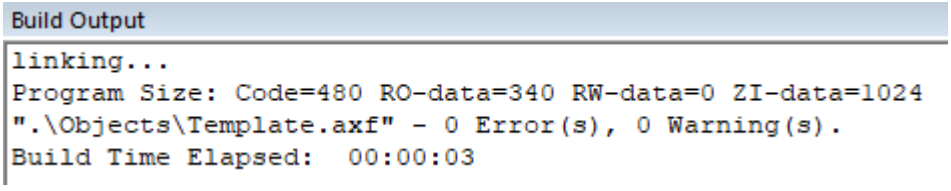
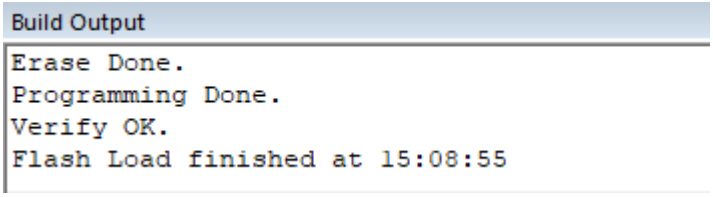


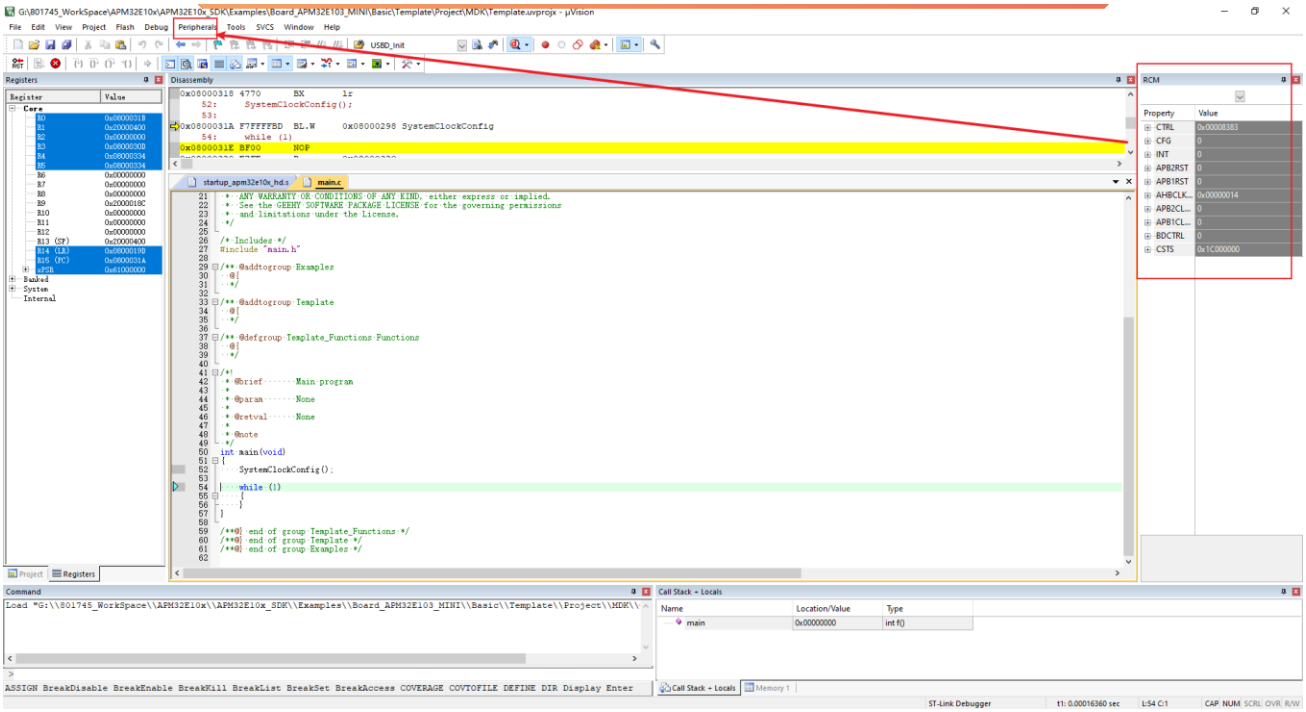
Figure 10 Download Results



10.4.3. Simulation Project

Click the "Start/Stop Debug Session" button on the interface to start simulation. On the simulation interface, you can use the "Peripherals", "Memory", and "Registers" tabs to view information such as the MCU core, peripheral registers, and memory data.

Figure 11 Simulation Project



11. Revision History

Table 5 Revision History

Date	Version	Revision History
September,2026	V1.0	New

12. Annex

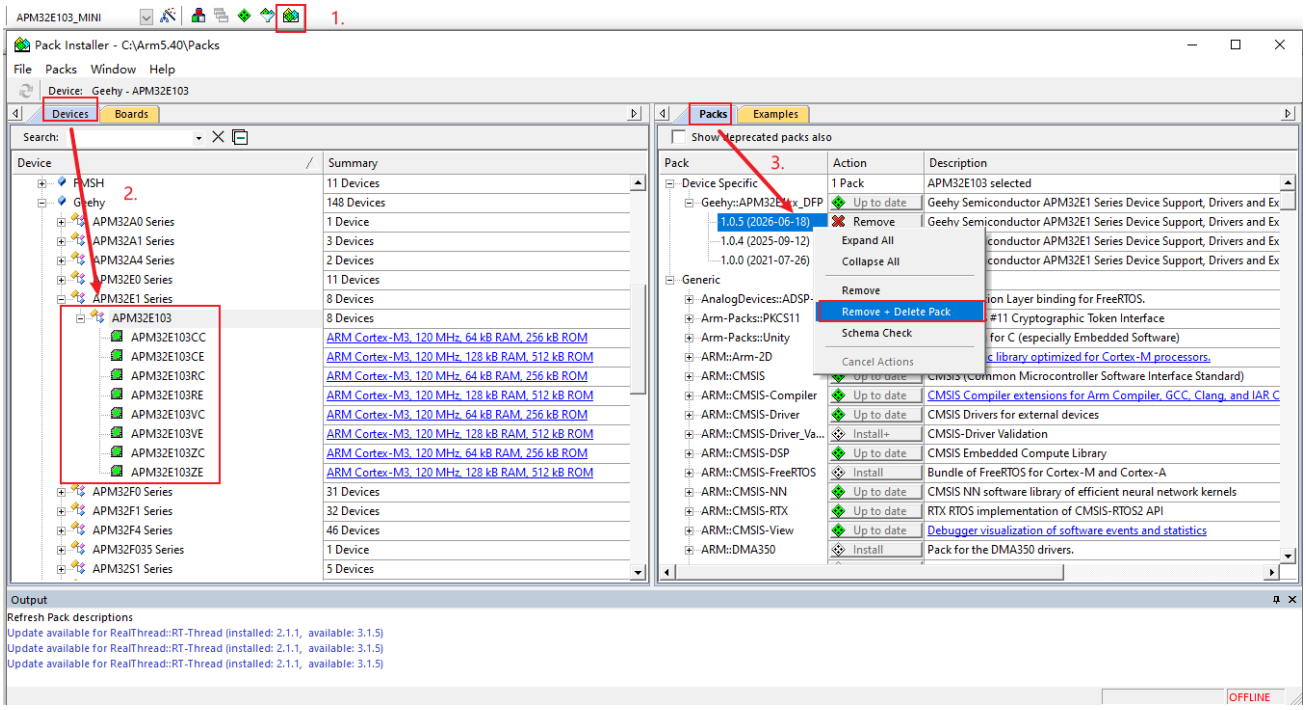
12.1. Uninstalling and Installing Keil pack

The following shows how to uninstall all versions of the pack and install the latest version of the pack.

Steps:

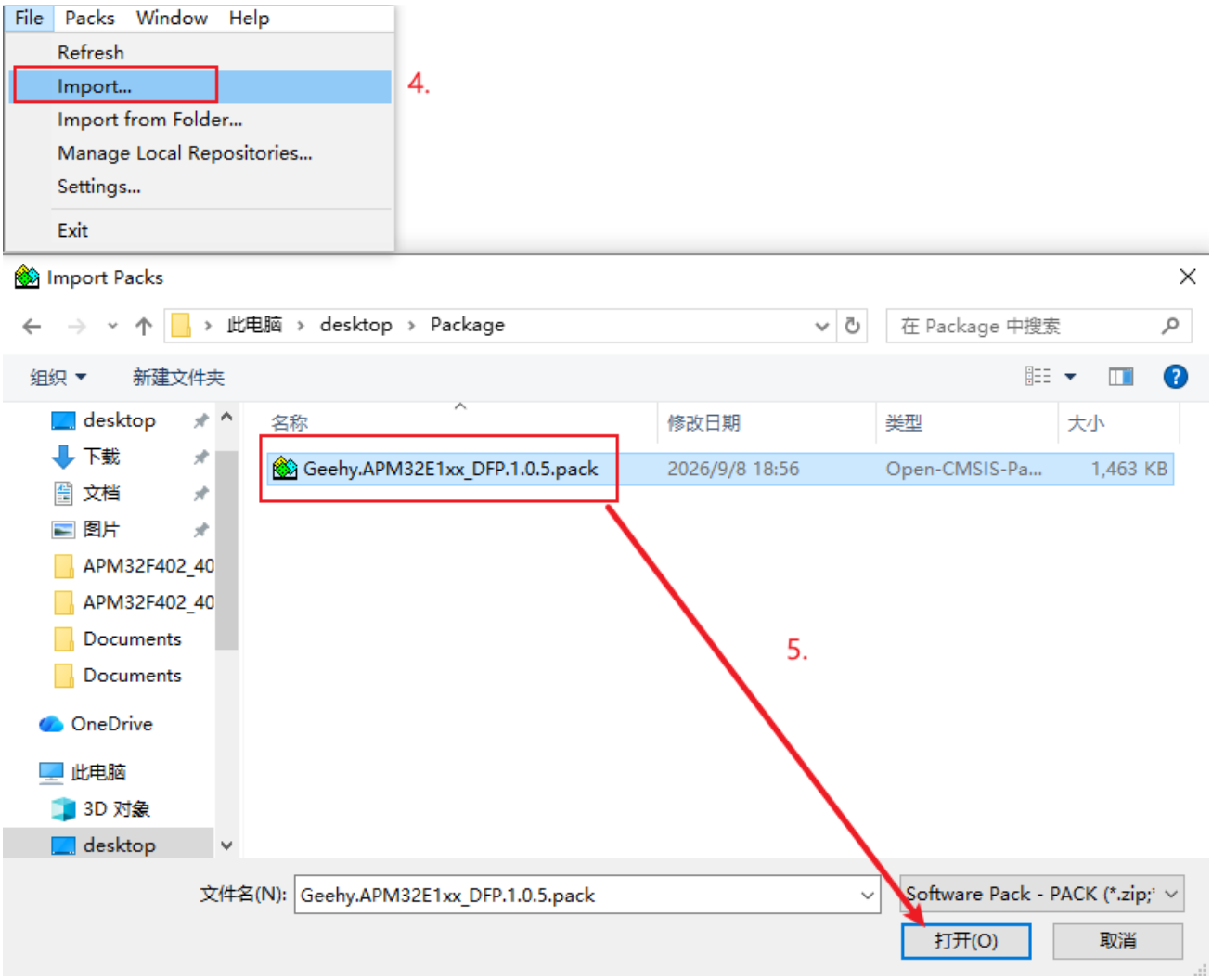
1. Open the pack installer.
2. In the Devices tab, select the corresponding chip model.
3. In the Packs tab, right-click and perform the delete operation.
4. In the pack installer, click File → Import to import and install the pack.
5. Select the pack and click Open to start the installation.
6. Check the installation directory of the pack to see if the pack contents have been updated.

Figure 12



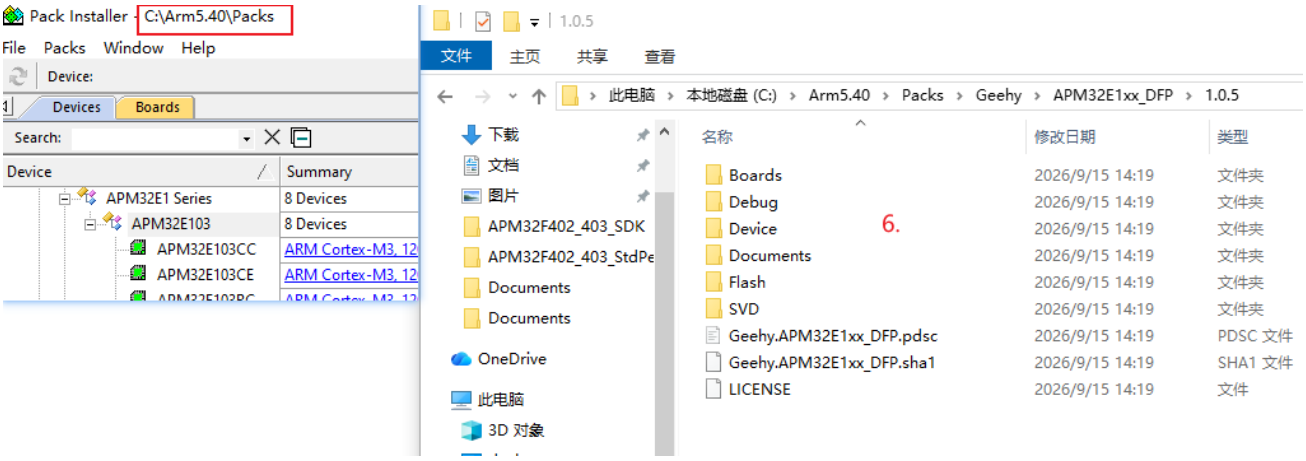
1. Open the pack installer
2. Select the chip model
3. Select the current version, right-click the mouse, and select “Remove + Delete Pack”

Figure 13



4. In the pack installer, select File → Import.
5. Select the pack and click Open for installation.

Figure 14



Pack Installer

6. Check the pack to see if it is the newest version.

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8. Scope of application

The information in this manual replaces the information provided in all previous versions of the manual.

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